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Supplementary material to Debt, Deleveraging and the Liquidity Trap

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February, 2012

Abstract

Here we summarize details and extensions of our basline model in Appendix A, B and C referred to in the main text of "Debt, Deleveraging and the Liquidity Trap."

The views expressed in the paper are those of the authors and are not necessarily reflective of views at the Federal Reserve Bank of New York or the Federal Reserve System. Any errors or omissions are the responsibility of the authors.

1 Appendix A (extended): Details on the model and the Approximation

This appendix summarizes the microfoundations of the simple general equilibrium model studied in the paper and shows how we obtain the log-linear approximations stated in the text. We also provide some more details relative to the Appendix reported with the main text.

1.1 Households

There is a continuum of households of mass 1 with χ^s of type s and $1-\chi^s$ of type b. Their problem is to maximize

$$E_0 \sum_{t=0}^{\infty} \beta^{(i)t} \left[u^i(C_t(i)) - v^i(h_t(i)) \right] \text{ where } i = s \text{ or } b$$

s.t.

$$B_t(i) = (1+i_{t-1})B_{t-1}(i) - W_t P_t h_t(i) - \int_0^1 \Pi_t(i) + P_t C_t(i) - T_t(i)$$

$$(1+r_t)\frac{B_t(i)}{P_t} \leq D_t(i)$$

where i_t is the nominal interest rate that is the return on one period riskfree nominal bond, while r_t is the riskfree real interest rate on a one period real bond. We derive the first order conditions of this problem by maximizing the Lagrangian

$$\begin{aligned}
\mathcal{L}_0(i) \quad = \quad & E_0 \sum_{t=0}^{\infty} \beta^{(i)t} \{ \left[u^i(C_t(i)) - v^i(h_t(i)) \right] \\
& + \phi_{1t}(i)[B_t(i) - (1+i_{t-1})B_{t-1}(i) + W_t P_t h_t(i) + \int_0^1 \Pi_t(i) - P_t C_t(i) - T_t(i)] \\
& + \phi_{2t}(i)[(1+r_t)\frac{B_t(i)}{P_t} - D_t(i)]\}
\end{aligned}$$

First order conditions

$$\frac{\partial \mathcal{L}_t(i)}{\partial C_t(i)} = u_c^i(C_t(i)) - \phi_{1t}(i)P_t = 0$$

$$\frac{\partial \mathcal{L}_t(i)}{\partial h_t(i)} = -v_h^i(h_t(i)) + P_t W_t \phi_{1t}(i) = 0$$

$$\frac{\partial \mathcal{L}_t(i)}{\partial B_t(i)} = \phi_{1t}(i) - \beta E_t \phi_{1t+1}(i)(1+i_t) + \phi_{2t}(i)\frac{(1+r_t)}{P_t} = 0$$

Complementary slackness condition

$$\phi_{2t}(i) \geq 0, \quad D_t(i) \geq (1+r_t)\frac{B_t(i)}{P_t}, \qquad \phi_{2t}(i)[(1+r_t)\frac{B_t(i)}{P_t} - D_t(i)] = 0$$

The $C_t(i)$ above refers to the Dixit-Stiglitz aggregator

$$C_t(i) = [\int_0^1 c_t(i,j)^{(\theta-1)/\theta} dj]^{\theta/(\theta-1)}$$

and P_t to the corresponding price index

$$P_t = [\int_0^1 p_t(j)^{(1-\theta)} dj]^{1/(\theta-1)}$$

The household maximization problem implies an aggregate demand function of good j given by

$$c_t(j) = C_t(\frac{p_t(j)}{P_t})^{-\theta}$$

1.2 Firms

There is a continuum of firms of measure one with a fraction λ the sets prices freely at all times and a fraction $(1-\lambda)$ that set their their prices one period in advance. Production is linear in labor $y_t(i) = h_t(i)$. The firms are held by the saver so we assume that profits are discounted by the marginal utility of income of the saver, ϕ_t^s, (this assumption plays no role in our log-linear economy but is stated for completeness):

$$E_t \sum_{t=0}^{\infty} \phi_t^s [(1-\tau) p_t(j) y_t(j) - W_t P_t h_t(j)]$$

s.t.

$$\begin{aligned} y_t(j) &= Y_t(\frac{p_t(j)}{P_t})^{-\theta} \\ y_t(j) &= h_t(j) \end{aligned}$$

Where τ is a monopoly wedge (set by policy) we have introduced for notational convenience (see in linearization). From this problem, we can see that the λ fraction of firms that set their price freely at all times they set their price so that

$$(1-\tau)\frac{p_t(1)}{P_t} = \frac{\theta}{\theta-1} W_t$$

and each charging the same price $p_t(1)$. Those that set their price one period in advance, however, satisfy

$$E_{t-1} \phi_t^s P_t^{1+\theta} Y_t p_t(2)^{-\theta-1} ((\frac{\theta-1}{\theta})\frac{p_t(2)}{P_t} - W_t) = 0$$

1.3 Government

Fiscal policy is the purchase of G_t of the Dixit-Stiglitz aggregate and the collects taxes T_t^s and T_t^b. The government's budget constraint is given by

$$\frac{B_t^g}{P_t} = (1+i_{t-1})\frac{P_{t-1}}{P_t}\frac{B_{t-1}^g}{P_{t-1}} + G_t - \chi_s T_t^s - \chi_b T_t^b$$

Note that the asset of the saver, will need to equal the debt of the government and the debtor so that

$$\chi_b B_t^b + B_t^g = -\chi_s B_t^s$$

For any variations in T_t^b or G_t we assume that current or future T_t^s will be adjusted to satisfy the government budget constraint, although we will relax this in section 7. Monetary policy is the choice of i_t. We assume it follows the Taylor rule specified in the text.

1.4 Steady state

In steady state the following relations hold for $(h^b, C^b, h^s, C^s, W, Y, \bar{r}, b^s)$ for exogenous values for $(b^g, \frac{T^s}{T^b}, G, D, \tau)$ where the latter set of variables need to satisfy the government budget constraint $\chi_s \bar{T}^s + \chi_b \bar{T}^b = \bar{G} - r\bar{b}^g$

$$\frac{v_h^b(\bar{h}^b)}{u_c^b(\bar{C}^b)} = W$$

$$\frac{v_h^s(\bar{h}^s)}{u_c^s(\bar{C}^s)} = W$$

$$W = (1-\tau)\frac{\theta - 1}{\theta}$$

$$\bar{C}^b = -r\bar{D} + \bar{W}\bar{h}^b - \bar{T}^b$$

$$\bar{Y} = \chi_s \bar{C}^s + \chi_b \bar{C}^b + \bar{G}$$

$$\chi_b \frac{\bar{D}}{1+r} + \bar{b}^g = -\chi_s \bar{b}^s$$

$$\bar{Y} = \chi_s \bar{h}^s + \chi_b \bar{h}^b$$

$$1 + \bar{r} = \beta^{-1}$$

We assume that τ is set to achieve the first best so that $(1-\tau)\frac{\theta-1}{\theta} = W = 1$. We consider a steady state of the model in which b borrows up to its limit, while the s does not, inflation is at zero (i.e. $\frac{P_t}{P_{t=1}} = 1$), $Y_t = \bar{Y}$ and $D_t = \bar{D}$. We parameterize the utility functions such that in equilibrium the steady state labor supply of the two types is the same, i.e. $\bar{h}^s = \bar{h}^b = h = \bar{Y}$.

1.5 Log linear approximation

Aggregate consumption is

$$C_t = \chi^s C_t^s + (1-\chi^s) C_t^b$$

where C_t^s and C_t^b is the of the consumption levels of each type. Similarly aggregate hours are

$$h_t = \chi^s h_t^s + (1-\chi^s) h_t^b$$

while aggregate output is given

$$Y_t = C_t + G_t$$

Lets start with linearizing the demand side. Assuming type b is up against his borrowing constraint and aggregating over all borrowers

$$C_t^b = -(\frac{1+i_{t-1}}{1+r_{t-1}})\frac{P_{t-1}}{P_t} D_{t-1} + \frac{1}{1+r_t}\frac{D_t}{P_t} + I_t^b - T_t^b$$

where I_t^b is the wage income of the borrower given by

$$I_t^b = W_t h_t^b$$

Log-linearizing this around $D_t = \bar{D}$ we obtain

$$\hat{C}_t^b = \hat{I}_t^b + \beta \hat{D}_t - \hat{D}_{t-1} + \gamma_D \pi_t - \gamma_D \beta (i_t - E_t \pi_{t+1} - \bar{r}) - \hat{T}_t^b$$

where $C_t^b \equiv \frac{C_t^b - \bar{C}^b}{\bar{Y}}$, $Y_t = \log Y_t / \bar{Y}$, $\hat{D}_t \equiv \frac{D_t - \bar{D}}{Y}$, i_t is now $\log(1 + i_t)$ in our previous notation, $\bar{r} \equiv \log \beta^{-1}$, $\pi_t \equiv \log P_t / P_{t-1}$, $T_t^b \equiv \frac{T_t^b - \bar{T}^b}{\bar{Y}}$, $\gamma_D = \frac{D}{\bar{Y}}$, $\hat{I}_t^b \equiv \frac{I_t^b - \bar{I}^b}{\bar{Y}}$ and

$$\hat{I}_t^b = \hat{W}_t + \hat{h}_t$$

For type s we obtain

$$u_c^s(C_t^s) = \beta(1 + i_t) E_t u_c^s(C_{t+1}^s) \frac{P_t}{P_{t+1}}$$

and log-linearizing this around steady state yields

$$\hat{C}_t^s = E_t \hat{C}_{t+1}^s - \sigma(i_t - E_t \pi_{t+1} - \bar{r})$$

where $C_t^s \equiv \frac{C_t^b - \bar{C}^s}{\bar{Y}}$ and $\sigma \equiv -\frac{u_c^s}{u_{cc}^s \bar{Y}}$. Aggregate consumption is then

$$\hat{C}_t = \chi^s \hat{C}_t^s + (1 - \chi^s) \hat{C}_t^b$$

where $C_t \equiv \log C_t / \bar{Y}$. and

$$\hat{Y}_t = \hat{C}_t + \hat{G}_t$$

where $\hat{Y}_t \equiv \frac{Y_t - \bar{Y}}{\bar{Y}}$, $\hat{G}_t \equiv \frac{G_t - \bar{G}}{\bar{Y}}$

Let us now turn to the production side. The pricing equations of the firms imply that

$$\log p_t(1) = \log P_t + \hat{W}_t$$

$$\log p_t(2) = \log P_t + \hat{W}_t$$

where $\hat{W}_t = \log W_t / \bar{W}$. This implies that

$$\log p_t(2) = E_{t-1} \log p_t(1)$$

Log linearizing the aggregate price index, implies

$$\log P_t = \lambda \log p_t(1) + (1 - \lambda) \log p_t(2)$$

so it follows (with a few manipulations using the equations above) that

$$\pi_t - E_{t-1} \pi_t = \log P_t - E_{t-1} \log P_{t-1} = \frac{\lambda}{1 - \lambda} [\log p_t(1) - \log P_t] = \frac{\lambda}{1 - \lambda} \hat{W}_t$$

To solve for W_t we linearize each of the optimal labor supply first order condition for each to yield

$$\begin{aligned} \hat{W}_t &= \omega^b \hat{h}_t^b + \sigma^{b-1} \hat{C}_t^b \\ \hat{W}_t &= \omega^s \hat{h}^s + \sigma^{s-1} \hat{C}_t^s \end{aligned}$$

where $\omega^b \equiv \frac{v^b_{hh}h}{v^b_h}, \omega^s \equiv \frac{v^s_{hh}Y}{v^s_h}$ and $\sigma^b \equiv -\frac{u^b_c}{u^b_{cc}Y}, \sigma^s \equiv -\frac{u^s_c}{u^s_{cc}Y}$ and $\hat{h}^b_t(i) \equiv \frac{h^b_t(i)-\bar{h}^b}{\bar{Y}}$ and $\hat{C}^b_t(i) \equiv \log\frac{C^b_t(i)-\bar{C}}{\bar{Y}}$.

Observe that $\hat{h}_t = \hat{Y}_t$. We now assume that $\omega^b = \omega^s = \omega$ and that $\sigma^b = \sigma^s = \sigma$. Using this we can now combine the labor supply of the two types to yield.

$$\hat{W}_t = \omega\hat{Y}_t + \sigma^{-1}\hat{C}_t$$

Combine this with our previous result, together with $\hat{Y}_t = \hat{C}_t + \hat{C}_t$ to yield

$$\pi_t = \frac{\lambda}{1-\lambda}(\omega + \sigma^{-1})\hat{Y}_t - \frac{\lambda}{1-\lambda}\sigma^{-1}\hat{G}_t + E_{t-1}\pi_t$$

$$\pi_t = \kappa\hat{Y}_t - \kappa\psi\hat{G}_t + E_{t-1}\pi_t$$

where $\kappa \equiv \frac{\lambda}{1-\lambda}(\omega+\sigma)$, $\psi \equiv \frac{\sigma^{-1}}{\sigma^{-1}+\omega}$.

2 Appendix B: Durable Goods and Investment dynamics

Here we extend the model to incorporate durable goods and productive investment.

2.1 Durable Consumption

2.1.1 Durable consumption in an endowment economy

Consider first and endowment economy in which each household receives Y_t in every period and everything is denominated in real terms. This endowment can be (i) invested in a "durable" consumption good via investment, $I_t(i)$ that depreciate at a rate δ and yields flow utility $d(K^i_t)$ or (ii) consumed at time t yielding flow utility $u(C_t)$. There is a continuum of households of mass 1 with χ^s of type s and $\chi_b = 1-\chi^s$ of type b. They only differ in the discount factor β_i where $i = s$ or b. Their problem is to maximize

$$E_0\sum_{t=0}^{\infty}\beta^t_i\left[u^i(C_t(i)) + d(K_t(i))\right] \text{ where } i = s \text{ or } b$$

s.t.

$$b_t(i) = (1+r_{t-1})b_{t-1}(i) - P_tY_t(i) + P_tC_t(i) + P_tI_t(i)$$

$$K_t(i) = I_t(i) + (1-\delta)K_{t-1}(i)$$

$$(1+r_t)b_t(i) \leq D_t$$

Substitute out for investment to get the Lagrangian

$$\begin{aligned}\mathcal{L}_0(i) \quad = \quad & E_0\sum_{t=0}^{\infty}\beta^{(i)t}\{u^i(C_t(i)) + d(K_t(i)) \\ & +\phi_{1t}(i)[b_t(i) - (1+r_{t-1})b_{t-1}(i) + P_tY_t(i) - P_tC_t(i) - P_t[K_t(i) - (1-\delta)K_{t-1}(i)] - T_t(i)] \\ & +\phi_{2t}(i)[(1+r_t)b_t(i) - D_t(i)]\}\end{aligned}$$

The first order conditions are

$$\frac{\partial\mathcal{L}_t(i)}{\partial C_t(i)} = u^i_c(C_t(i)) - \phi_{1t}(i) = 0$$

$$\frac{\partial \mathcal{L}_t(i)}{\partial K_t(i)} = d_K^i(CK_t(i)) - \phi_{1t}(i) + \beta\phi_{1t+1}(i)(1-\delta) = 0$$

$$\frac{\partial \mathcal{L}_t(i)}{\partial b_t(i)} = \phi_{1t}(i) - \beta E_t\phi_{1t+1}(i)(1+r_t) + \phi_{2t}(i)(1+r_t) = 0$$

The complementary slackness conditions are

$$\phi_{2t}(i) \geq 0, \quad D_t \geq (1+r_t)b_t(i), \qquad \phi_{2t}(i)[(1+r_t)D_t(i) - D^i] = 0$$

Substituting out for the lagrange multipliers we can summarize the model in the following set of equations (we do not report the inequality constraints but will be specific about D_t^b below.

$$u_c^s(C_t^s) = \beta_s(1+r_t)E_t u_c^s(C_{t+1}^s)$$

$$d_K^b(K_t^b) - u_c^b(C_t^b) + (1-\delta)\beta_b u_c^b(C_{t+1}^b) = 0$$

$$d_K^s(K_t^s) - u_c^b(C_t^s) + (1-\delta)\beta_s u_c^s(C_{t+1}^s) = 0$$

$$Y_t = C_t + I_t$$

$$K_t = I_t + (1-\delta)K_{t-1}$$

$$C_t = \chi_s C_t^s + \chi_b C_t^b$$

$$I_t = \chi_s I_t^s + \chi_b I_t^b$$

$$b_t^b = (1+r_{t-1})b_{t-1}^b - Y_t + C_t^b(i) + I_t^b(i)$$

Relative to our earlier treatment what is new here is the arbitrage equation between the durable goods and the perishable good.

The steady state of this economy, given $(\bar{Y}, \bar{D})$, is a set of constants $(r, \bar{I}^b, \bar{I}^s, \bar{K}^s, \bar{K}^b, \bar{C}^b, \bar{C}^s)$ that solve the following equations:

$$\frac{d_K^b(\bar{K}^b)}{u_c^b(\bar{C}^b)} = 1 - \beta_b(1-\delta)$$

$$\frac{d_K^s(\bar{K}^s)}{u_c^s(\bar{C}^s)} = 1 - \beta_s(1-\delta)$$

$$\frac{\bar{K}^s}{\bar{I}^s} = \delta$$

$$\frac{\bar{K}^b}{\bar{I}^s} = \delta$$

$$\bar{r} = \beta_s^{-1} - 1$$

$$\frac{\bar{r}}{1+\bar{r}}\bar{D} = \bar{Y} - \bar{C}^b - \bar{I}^b$$

$$\bar{Y} = \chi_b \bar{C}^b + \chi_s \bar{C}^s + \chi_b \bar{I}^b + \chi_s \bar{I}^s$$

The steady state value of durable goods here will be determined the preference the household has for spending on durables vs. nondurables.

We now linearize the model around $(\bar{Y}, \bar{D})$ and $(\bar{I}^b, \bar{I}^s, \bar{K}^s, \bar{K}^b, \bar{C}^b, \bar{C}^s)$. Let us denote the steady state ratios $\frac{i}{Y}$ by γ_i where i can be any of the steady state variables. Observe that we will be assuming, as in the paper that the path for the debt limit is $D_{-1} = D^{high,}$ and $D_t = D^{low} = \bar{D}$ for $t \geq 0$. Furthermore let us define $\sigma^b = -\frac{u_c^b}{u_{cc}^b C^b}$ and $\sigma^s = -\frac{u_c^s}{u_{cc}^s C^s}$ and $\vartheta^b = -\frac{d_K^b}{d_{KK}^b K^b}$, $\vartheta^s = -\frac{d_{KK}^s K^s}{d_K^s}$.

Then we can write the model in its linearized form as

$$\hat{C}_t^s = \vartheta_s^{-1}\sigma_s[1-\beta_s(1-\delta)]\hat{K}_t^s + \beta_s(1-\delta)E_t\hat{C}_{t+1}^s \tag{1}$$

$$\hat{C}_t^b = \vartheta_b^{-1}\sigma_b[1-\beta_b(1-\delta)]\hat{K}_t^s + \beta_b(1-\delta)E_t\hat{C}_{t+1}^b \tag{2}$$

$$\hat{K}_t^s = \delta\hat{I}_t^s + (1-\delta)\hat{K}_{t-1}^s \tag{3}$$

$$\hat{K}_t^b = \delta\hat{I}_t^b + (1-\delta)\hat{K}_{t-1}^b \tag{4}$$

$$\hat{Y}_t = \chi_s\gamma_{C^s}\hat{C}_t^s + \chi_b\gamma_{C^b}\hat{C}_t^b + \chi_s\gamma_{I^s}\hat{I}_t^s + \chi_b\gamma_{I^b}\hat{I}_t^b \tag{5}$$

$$\hat{C}_t^s = E_t\hat{C}_{t+1}^s - \sigma_s\hat{r}_t \tag{6}$$

$$-\gamma_D\beta_s\hat{r}_t = -\hat{Y}_t + \gamma_{C_b}\hat{C}_t^b + \gamma_{I_b}\hat{I}_t^b + \gamma_D\hat{D}_{t-1} \tag{7}$$

where $\hat{D}_t = \frac{D_t - D^l}{D^l}$, $\hat{C}_t^s = \frac{C_t^s - C^s}{C^s}$, $\hat{K}_t^s = \frac{K_t^s - K^s}{K^s}$, $\hat{I}_t^s = \frac{I_t^s - I^s}{I^s}, \hat{r}_t = \frac{r_t - \bar{r}}{1+\bar{r}}$ and we define the variables symmetrically for the borrower. Our assumption about the path for D_t implies that $\hat{D}_{-1} > 0$ but $\hat{D}_t = 0$ for all $t \geq 0$

Recall that there is an unexpected reduction in the debt limit at period zero, $D^h -> D^l$, and we are linearizing around the new lower level of this debt limit. From the steady state relationships above we can derive

$\chi_b\gamma_{C^b} + \chi_b\gamma_{I^b} = \chi_b - \chi_b\bar{r}\gamma_D$
$\chi_s\gamma_{C^s} + \chi_s\gamma_{I^s} = \chi_s + \chi_b\bar{r}\gamma_D$
$\gamma_{I^s} + \gamma_{I^b} = \frac{\text{Durable consumption}}{\text{Total consumption}}$

We make choose a number for the ratios $\frac{\text{Durable consumption}}{\text{Total consumption}}$ and γ_D and then we need one additional assumption to fully parameterize the model, given $\beta_s, \beta_b, \chi_s, \chi_{b,}$ and δ. We simply assume that utility is such that in steady state the borrowers and the savers spend the same fraction of their income on durables versus non-durables which implies that

$$\frac{\gamma_{C^b}}{\gamma_{I^b}} = \frac{\gamma_{C^s}}{\gamma_{I^s}}$$

which allows us to back out all the $\gamma_i's$ with the formulas

$$\gamma_{I^b} = \frac{(1-\bar{r}\gamma_D)}{(2+[\chi_s^{-1}(\chi_b-\chi_s)]\bar{r}\gamma_D)}\frac{\text{Durable consumption}}{\text{Total consumption}}$$

$$\gamma_{I^s} = \frac{\text{Durable consumption}}{\text{Total consumption}} - \gamma_{I^b}$$

etc. What remains is to choose the value of σ_s, σ_b and ϑ_s, ϑ_b.

As in the body of the paper we assume that $\sigma = \sigma^s\gamma_{C_s} = \sigma^b\gamma_{C_b}$ and we also assume that $\vartheta = \vartheta^s\gamma_{C_s} = \vartheta^b\gamma_{C_b}$ which implies that $\sigma\vartheta^{-1} = \vartheta_s^{-1}\sigma_s = \vartheta_b^{-1}\sigma_b$. Observe that if, under this assumption, we choose σ^s then we can back out the other parameters. The table below summarizes the parameters of the model. We assume the short run is "two years". Hence the ratio γ_D which is level of debt over output (which is two year production) being 0.5, means that the household debt is 100 percent of annual output.;

Table 1: Numerical example

parameters	values	parameters	values
β_s	0.98	δ (annual)	0.25
β_b	0.84	γ_D	0.5
σ^s	1	$\hat{D}_{-1}$	0.3
ϑ^s	1	Model with endogenous production	
χ_s	0.66	ω	1
$\frac{\text{Durable consumption}}{\text{Total consumption}}$	0.1	λ	0.5

2.1.2 Durable consumption with production, sticky prices and the zero bound

We now extend the model with durable consumption to include endogenous production. The steps here are the same as in the model in the paper, hence, we only report the linearized model. Again, equation (1)-(6) apply, with the real rate given by

$$\hat{r}_t = i_t - \bar{r} - E_t\pi_{t+1} \tag{8}$$

and the central bank set policy according to

$$i_t = \max(0, \bar{r} + \phi_\pi\pi_t) \tag{9}$$

We now replace 7 with

$$\gamma_{C_b}\hat{C}_t^b + \gamma_{I_b}\hat{I}_t^b = \mu\hat{Y}_t - \gamma_D\hat{D}_{t-1} + \gamma_D\pi_t - [\gamma_D\beta + \omega^{-1}\chi_b^{-1}\chi_s]\hat{r}_t \tag{10}$$

where $\mu = (1+\omega^{-1})(\omega+\sigma^{-1}) - \sigma^{-1}\omega^{-1}\chi_b^{-1}$. Recall that that $\hat{D}_{-1} > 0$ but $\hat{D}_t = 0$ for all $t \geq 0$. What remains, then, is to add New Classical Phillips curve which is given by

$$\pi_t = \kappa\hat{Y}_t + E_{t-1}\pi_t \tag{11}$$

where $\kappa = \frac{\lambda}{1-\lambda}(\omega+\sigma^{-1})$. The model, then, is characterized by (1)-(6) and (8)-(11). In order to finish parameterizing the model, we need to specify ω, which measures the curvature of the disutility of labor, and λ, which is the fraction of firms that preset their prices, and the level of debt overhang, $\hat{D}_{-1}$, see Table 1.

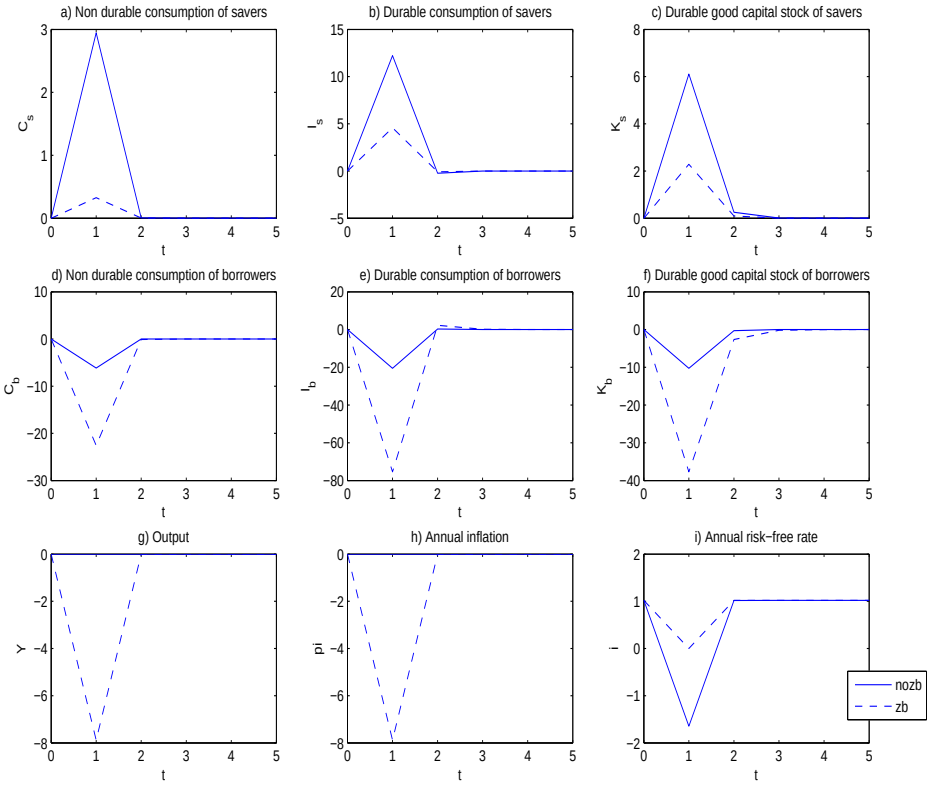


Figure 1: The model with durable goods

2.1.3 Results

The results are shown in figure 1 and table 2. The point of the exercise is that if a certain fraction of goods are more "durable", then the deleveraging consumer will cut down his spending on durables more than on non-durables. In the baseline parameterization we assume that the 25 percent of the durable capital depreciates per year, the rest of the parameters are documented in Table 1. The solid line in the figure reflects the solution of the model if prices were flexible (or alternatively if the zero bound is not imposed and the central bank targets zero inflation) while the dashed line shows the solution when the zero bound is imposed. As the figure reveals, the borrower will cut back on spending on durables by about three times more than he cuts down on spending on durables, and this is true under both sticky and flexible prices. The logic is straight forward: If a borrower needs to temporarily cut down on spending in the short run to satisfy a borrowing limit, it makes sense to do so by spending less on those goods that have a long duration, since the payoff from spending on durables will extend over a long period of time. This insight applies whether or not we impose the zero bound or sticky prices. The main difference in assuming sticky prices and imposing the zero bound is evident in aggregate output and inflation. Under flexible prices, the saver will makes up for the drop in spending by increasing consumption of both durables and non-durables, leaving aggregate output unchanged. The way in which the saver is induced to spend is via a reduction in the real interest rate, as seen in the figure. Price frictions and the zero bound, however, prevent this adjustment to take place and hence the increase in consumption by the saver is no enough to make up for the drop in spending by the borrower. As a consequence output (and inflation) drop in the short run. (For a multiperiod deleveraging process, then consumption of the saver may even drop if people expect deflation which means the real interest rate will increase).

Column1	Column2	δ=1	Column3	δ=0.75	Column4	δ=0.5	Column5	δ=0.25	Column6	δ=0.1
	Sticky	Flexible	Sticky	Flexible	Sticky	Flexible	Sticky	Flexible	Sticy	Flexible
C^s_S	2.0	3.3	1.8	3.3	1.4	3.2	0.3	2.9	-2.2	2.5
I^s_S	2.0	3.3	2.5	4.4	3.2	6.4	4.4	12.2	6.2	27.4
C^b_S	-20.5	-6.7	-20.8	-6.6	-21.4	-6.5	-23.4	-6.1	-26.5	-5.8
I^b_S	-20.5	-6.7	-27.0	-8.6	-39.9	-12.1	-78.1	-20.5	-176.9	-39.0
Y_S	-5.4	0.0	-5.7	0.0	-6.3	0.0	-8.2	0.0	-12.3	0.0
π_S	-5.5	0.0	-5.8	0.0	-6.5	0.0	-8.5	0.0	-12.6	0.0

That consumers will deleverage most by cutting spending on durables is particularly clear in table 2 which show the percentage change in each variable in the short run. It compares spending on the non-durable good (C^b_t) to the spending on the durable good (I^b_t) under different degree of depreciation of the durable good. $\delta = 1$ means that the durable good and the non-durable goods are exactly the same and all goods are non-durable. As the table suggests then the borrower deleverages in the short-run by cutting spending on each component by exactly the same amount. As we reduce the depreciation of the durable good we see that the borrower will start cutting spending on the durables by more and more in the short run. For depreciation of 25 percent, for example, spending on durable goods collapses by 75 percent, while assuming even lower depreciation can imply that the borrower starts "eating" his durables.

2.2 Productive Investment

2.2.1 Investment with flexible prices

We now consider the effect of introducing capital goods that have no effect on utility but enter as a factor of production. The households can hold capital and rent to the firms that produce at a rental price of q_t. Firms that have a constant return to scale production function of the form:

$$y_t = k_t^{\alpha} l_t^{1-\alpha}$$

To motivate borrowing and lending we assume that one of the household types (the borrowers) can investment in capital, while the other type (the saver) can only invest in a one period risk-free bond. In order to have borrowing and lending in steady state we need to ensure that the borrowers cannot self-finance in the long run (see, e.g., discussion in Bernanke, Gertler and Gilchrist (1999)). A common way of doing this is to give the borrower a finite horizon, e.g. due to exogenous entries and exits. A mathematical equivalent way to accomplishing this is to simply assume the borrower is more "impatient", a short-cut we use here for a better comparison with the other variations of the model in the paper.

We assume that there is an adjustment cost of investment given by $\gamma(\frac{I_t^b}{\bar{I}})$ with $\gamma(\frac{\bar{I}}{\bar{I}}) = \gamma_I(\frac{\bar{I}}{\bar{I}}) = 0$ and $\gamma_{II}(\frac{\bar{I}}{\bar{I}}) > 0$. The reason we introduce the adjustment cost is that in this way we approximate our previous specification in the main text. If the adjustment cost is very high, then the model is equivalent to a model with a model with a fixed capital stock. The only difference with the model in the main text, then, is that there is a capital income that accrues to the borrower that does not show up in the model without capital.

The maximization problem of the borrower is to maximize

$$E_0 \sum_{t=0}^{\infty} \beta_b^t \left[[u^b(C_t^b) - v^b(h_t^b)] \right]$$

s.t.

$$b_t^b = (1 + r_{t-1}) b_{t-1}^b - w_t h_t^b - q_t K_t^b + I_t^b + C_t^b + T_t + \gamma(\frac{I_t^b}{\bar{I}})$$

$$K_t^b = I_t^b + (1 - \delta) K_{t-1}^b$$

$$(1 + r_t) b_t^b \leq D^{high/low}$$

The Lagrangian for this problem is

$$\begin{aligned} \mathcal{L}_0^b \quad = \quad & E_0 \beta_b^t \{ u^b(C_t) - v^b(h_t)) \\ & + \phi_{1t} [b_t^b - (1 + r_{t-1}) b_{t-1}^b + w_t h_t^b + q_t K_t - (K_t - (1 - \delta) K_{t-1}) - C_t^b - T_t^b - \gamma(\frac{K_t - (1 - \delta) K_{t-1}}{\bar{I}}] \\ & + \phi_{2t} [(1 + r_t) b_t^b - D^{high/low}] \} \end{aligned}$$

yielding the first order conditions

$$\frac{\partial \mathcal{L}_0}{\partial C_t} = u_{c,t}^b - \phi_{1t} = 0$$

$$\frac{\partial \mathcal{L}_0}{\partial h_t} = v_{h,t}^b + \phi_{1t} w_t = 0$$

$$\frac{\partial \mathcal{L}_0}{\partial K_t} = (q_t - 1 - \gamma_K(\frac{I_t}{\bar{I}})) \phi_{1t} + \beta_b [(1 - \delta)(1 + \gamma_K(\frac{I_{t+1}}{\bar{I}}))] E_t \phi_{1t+1} = 0$$

$$\frac{\partial \mathcal{L}_0}{\partial B_t} = \phi_{1t} + \beta_b(1+r_t)E_t\phi_{1t+1} + \phi_{2t} = 0$$

and a complementary slackness condition

$$\phi_{2t} \geq 0, \quad \phi_{2t}[(1+r_t)D_t^b - D^{\text{high/low}}] = 0$$

The maximization problem for the saver is

$$E_0 \sum_{t=0}^{\infty} \beta_s^t \left[[u^s(C_t^s) - v^s(h_t^s)\right]$$

s.t.

$$D_t^s = (1+r_{t-1})D_{t-1}^s - w_t h_t^s + C_t^s + T_t^s$$

$$(1+r_t)D_t^s \leq D^{\text{high/low}}$$

The Lagrangian for this problem is

$$\mathcal{L}_0^s = E_0\beta_s^t(u^s(C_t) - v^s(h_t)) + \psi_{1t}[b_t^s - (1+r_{t-1})b_{t-1}^s + w_t h_t^s - C_t^s - T_t^s] + \psi_{2t}[(1+r_t)b_t^b - D^{\text{high/low}}]$$

$$\frac{\partial \mathcal{L}_0^s}{\partial C_t^s} = u_c^s - \psi_{1t} = 0$$

$$\frac{\partial \mathcal{L}_0^s}{\partial h_t^s} = v_h^s + \psi_{1t}w_t = 0$$

$$\frac{\partial \mathcal{L}_0^s}{\partial D_t^s} = \psi_{1t} - \gamma(1+r_t)E_t\psi_{1t+1} + \psi_{2t} = 0$$

and a complementary slackness condition

$$\psi_{2t} \geq 0, \quad \psi_{2t}[(1+r_t)b_t^b - D^{high/low}] = 0$$

We assume competitive factor markets, hence, given that firms have a constant return to scale, their capital labor ratios satisfy

$$\alpha(\frac{k_t}{l_t})^{\alpha-1} = q_t$$

$$(1-\alpha)(\frac{k_t}{l_t})^{\alpha} = w_t$$

We now impose market clearing and can then summarize the model (by substituting out the lagrange multipliers and impose market clearing) with the following equations (again we do not report the inequality constraints in this summary, but will make the same assumption as in earlier sections)

$$(1+\gamma_K(\frac{I_t}{\bar{I}}) - q_t)u_c^b(C_t^b) = \beta_b(1-\delta)(1+\gamma_K(\frac{I_{t+1}}{\bar{I}}))E_t u_c^b(C_{t+1}^b)$$

$$u_c^s(C_t^s) = \beta_s(1+r_t)E_t u_c^s(C_{t+1}^s)$$

$$\frac{v_h^s(\bar{h}_t^s)}{u_c^s(\bar{C}_t^s)} = w_t$$

$$\frac{v_h^b(\bar{h}_t^b)}{u_c^b(\bar{C}_t^b)} = w_t$$

$$Y_t = (\chi_b K_t)^\alpha L_t^{1-\alpha}$$

$$(1-\alpha)(\chi_b K_t)^\alpha L_t^{-\alpha} = w_t$$

$$\alpha(\chi_b K_t^b)^{\alpha-1} L_t^{1-\alpha} = q_t$$

$$L_t = \chi_s h_t^s + \chi_b h_t^b$$

$$C_t = \chi_s C_t^s + \chi_b C_t^b$$

$$b_t^b = (1+r_{t-1})b_{t-1}^b - w_t h_t^b - q_t K_t^b + I_t^b + C_t^b + T_t^b$$

$$K_t^b = I_t^b + (1-\delta)K_{t-1}^b$$

$$Y_t = C_t + I_t + \gamma(\frac{I_t}{I}) = C_t + \chi^b(K_t^b - (1-\delta)K_{t-1}^b) + \gamma(\frac{I_t}{I})$$

Steady state: As before we make the assumption that in steady state $h^b = h^s = L$. In steady state we have from the two euler equations that

$$r = \beta_s^{-1} - 1$$

and

$$q = 1 - \beta_b(1-\delta)$$

The Cobb-Douglas production function and competitive factor markets imply that $\frac{wL}{Y} = 1-\alpha$ and $\frac{q\chi_b K^b}{Y} = \alpha$. Let us denote the steady state ratios $\frac{i}{Y}$ by γ_i where i can be any of the steady state variables but these ratios will show up in the linearization. First we choose some value for γ_D which is exogenously given in the model. Using the expressions above, i.e. the steady state value of q and steady state capital output ratio, we have

$$\gamma_K = \frac{\alpha}{1-\beta_b(1-\delta)}$$

and

$$\gamma_C = 1 - \delta\gamma_K$$

We can use the savers budget constraint to show that

$$\gamma_{C^s} = 1 - \alpha + r\gamma_D$$

and use $\gamma_C = \chi_b\gamma_{C^b} + \chi_s\gamma_{C_s}$ to back out the implied value

$$\gamma_{C^b} = (\gamma_C - \chi_s\gamma_{C_s})\chi_b^{-1}$$

Finally we need to make assumption about γ_{T^b} depending on the fiscal policy regime considered.

We now linearize the model around $(\bar{Y}, \bar{D})$ and $(\bar{I}^b, \bar{I}^s, \bar{K}^s, \bar{K}^b, \bar{C}^b, \bar{C}^s)$. Observe that we will be assuming, as in the paper that the path for the debt limit is $D_{-1} = D^{high,}$ and $D_t = D^{low} = \bar{D}$ for $t \geq 0$. Furthermore let us define $\sigma^b = -\frac{u_c^b}{u_{cc}^b C^b}$ and $\sigma^s = -\frac{u_c^s}{u_{cc}^s C^s}$ and $\vartheta^b = -\frac{d_K^b}{d_{KK}^b K^b}$, $\vartheta^s = -\frac{d_{KK}^s K^s}{d_K^s}$, $\chi_q \equiv \frac{q}{1-q} = \frac{1-\beta_b\delta(1-\delta)}{\beta_b(1-\delta)}$

Then we can write the model in its linearized form as

$$\hat{C}_t^b - \frac{\sigma^b \gamma_{II}}{\beta_b(1-\delta)}\hat{I}_t + \sigma^b \chi_q \hat{q}_t = E_t \hat{C}_{t+1}^b - \sigma^b \gamma_{II} E_t \hat{I}_{t+1} \tag{12}$$

$$\hat{C}_t^s + \sigma^s \hat{r}_t = E_t \hat{C}_{t+1}^s \tag{13}$$

$$\hat{w}_t = \sigma_s^{-1}\hat{C}_t^s + \omega^s \hat{h}_t^s \tag{14}$$

$$\hat{w}_t = \sigma_b^{-1}\hat{C}_t^b + \omega^b \hat{h}_t^b \tag{15}$$

$$\hat{Y}_t = \alpha \hat{K}_t + (1-\alpha)\hat{L}_t \tag{16}$$

$$\alpha \hat{K}_t - \alpha \hat{L}_t = \hat{w}_t \tag{17}$$

$$-(1-\alpha)\hat{K}_t + (1-\alpha)\hat{L}_t = \hat{q}_t \tag{18}$$

$$L_t = \chi_s h_t^s + \chi_b h_t^b \tag{19}$$

$$\gamma_C \hat{C}_t = \chi_s \gamma_{C^s} \hat{C}_t^s + \chi_b \gamma_{C^b} \hat{C}_t^b \tag{20}$$

$$\hat{Y}_t = \gamma_C \hat{C}_t + \gamma_K \hat{K}_t - (1-\delta)\gamma_K \hat{K}_{t-1} \tag{21}$$

$$\hat{K}_t = \delta \hat{I}_t + (1-\delta)\hat{K}_{t-1}$$

$$\gamma_{C^b}\hat{C}_t^b = -\beta_s \gamma_D \hat{r}_t - \gamma_D \hat{D}_{t-1} + \gamma_D \pi_t + (1-\alpha)\hat{w}_t + (1-\alpha)\hat{h}_t^b + \alpha\chi_b^{-1}\hat{K}_t + \alpha\chi_b^{-1}\hat{q}_t - \gamma_K \hat{K}_t + (1-\delta)\gamma_K \hat{K}_{t-1} - \gamma_{T^b}\hat{T}_t^b \tag{22}$$

where again $\hat{D}_{-1} > 0$ but $\hat{D}_t = 0$ for all $t \geq 0$.

2.2.2 Investment with sticky prices and the zero bound

We now extend the model to incorporate sticky prices and the zero bound. Again the steps are standard, and more or less the same as in the paper and our previous section (note that the common assumption that the firms can rent labor and capital from in a economy wide market without any adjustment costs simplifies things a bit, as this means that the capital/labor ratio across firms is the same, and also that marginal costs across firms are equalized). The real interest rate is now

$$\hat{r}_t = i_t - \bar{r} - E_t \pi_{t+1} \tag{23}$$

and the central bank set policy according to

$$i_t = \max(0, \bar{r} + \phi_\pi \pi_t) \tag{24}$$

What remains, then, is to add New Classical Phillips curve which is given by

$$\pi_t = \frac{\lambda}{1-\lambda}(1-\alpha)\hat{w}_t + \frac{\lambda}{1-\lambda}\alpha\hat{q}_t + E_{t-1}\pi_t \tag{25}$$

The model, then, is characterized by (12)-(25). In order to parameterizing the model, we need to choose the same parameters as before, but instead of the ratio between durable goods and total consumption, we pick the share of capital in the production function, i.e. α. Other parameters are chosen as before with one exception, see table. We increase the share of savers from 0.66 to 0.7 when $\gamma_{II} = \infty$, and to 0.9 when $\gamma_{II} = 0.5$. By increasing the number of savers we are reducing the effect of the deleveraging shock. This reflects that demand is dropping more as one introduces flexible investment (we choose a higher number of savers so as not to exaggerate the drop in output relative to the previous example and make them better comparable).

parameters	values	parameters	values
β_s	0.98	δ (annual)	0.25
β_b	0.84	γ_D	0.5
σ^s	1	$\hat{D}_{-1}$	0.3
ϑ^s	1	ϕ_{II}	$0.45, \infty$
χ_s	$0.725; 0.9$	ω	1
α	0.3	λ	0.5

2.2.3 Results

Figure 2 and the table below shows the results with productive capital. It compares the model with infinite adjustment costs $\gamma_{II} = \infty$ to one in which they are finite. As the the adjustment costs become smaller, the drop in demand becomes bigger. Recall that to make the two figures better comparable we have increased the fraction of savers from 0.7 with infinite adjustment costs to 0.9 with adjustment costs so that the output drop is comparable across the two numerical experiments. The main difference, then, is in the composition. Once the borrowers can deleverage by cutting productive investment they will do so more aggressively so that the drop in aggregate investment is several times bigger than the drop in aggregate consumption. This is broadly consistent with the empirical evidence during the Great Recession and the Great Depression.

Column1	Column2	Column3
	γ_{II}=Inf	γ_{II}=0.5
	χ_S=0.7	χ_S=0.9
Y_S	-7.62	-8.57
C_S	-10.28	-3.98
I_S	0	-21.73
π_S	-8.13	-3.3
r_S	0	0

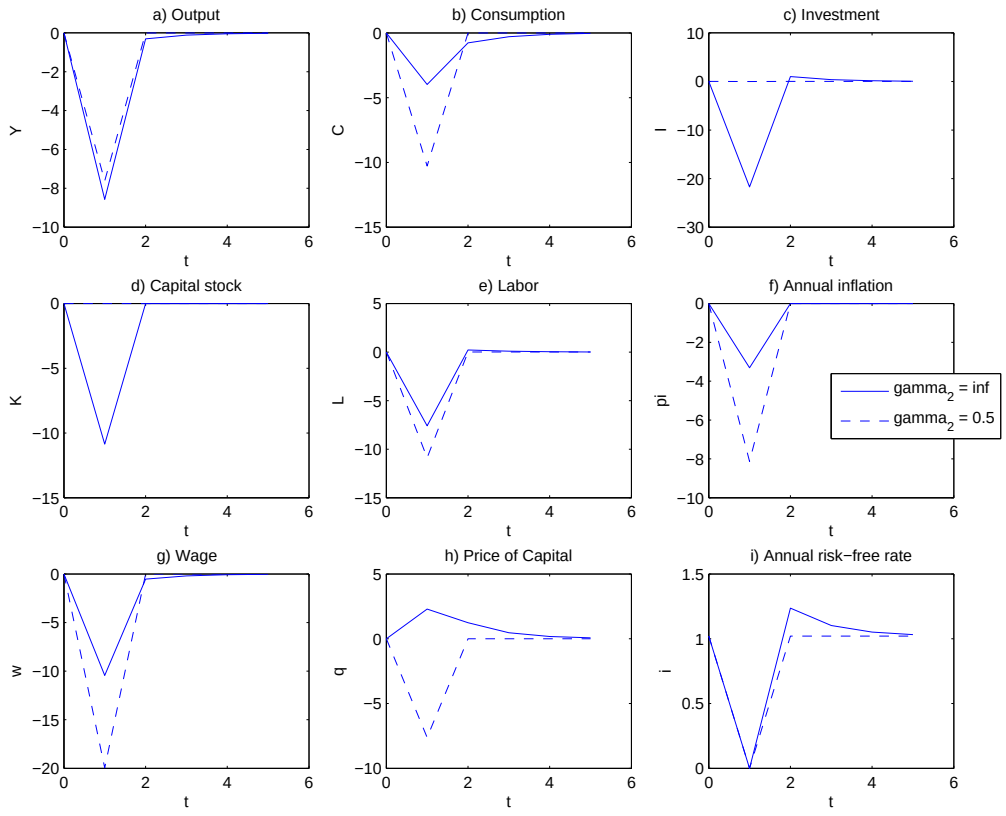


Figure 2: The model with productive capital

3 Appendix C: Dynamic Deleverging

Here we extend the model to incorporate dynamic deleveraging.

3.1 Households

We start with households. Borrowers are of measure $\chi_b = \chi$ and maximize

$$E_0 \sum_{t=0}^{\infty} \delta^t [u^b(C_t^b) - v^b\left(h_t^b\right)]$$

s.t.

$$B_t = (1 + i_{t-1}^b)B_{t-1} + P_t C_t^b - P_t W_t h_t^b - \int I_t^b(i)di - \int Z^b(i)di - P_t F_t^b + P_t T_t^b$$

where a positive B_t means a debt, i_t^b is the nominal interest rate on borrowing from banks, C_t^b is the consumption of borrowers, h_t^b labor, W_t real wages, δ is the discount factor and $0 < \delta < 1$. Here I_t^b is profits from the financial intermediaries distributed to the borrowers, $Z_t^b(i)$ is profits from the retail firms, F_t^b is revenues from "fraud" while T_t^b is taxes (fraud will be explained in the special section on "banking").

Lenders are of measure $\chi_s = 1 - \chi$

$$E_0 \sum_{t=0}^{\infty} \beta^t [u^s(C_t^s) - v^s\left(h_t^s\right)]$$

$$D_t = (1 + i_t^d)D_{t-1} - P_t C_t^s + P_t W_t h_t^s + \int I_t^s(i)di + \int Z_t^s(i)di + P_t F_t^b - P_t T_t^s$$

where a positive D_t means deposits, i_t^d is the interest rate the household receives on deposits it has at "banks", the rest of the notation is symmetric to the borrower above and here β is the discount factor that satisfies $0 < \delta < \beta < 1$.

$$1 + i_t^b = (1 + i_t^d)(1 + \omega_t) \tag{26}$$

where ω_t is given by the function

$$\omega_t = \omega(\frac{B_t}{P_t}, \frac{M_t}{P_t}, b^j) \; j = H, L \tag{27}$$

where $\frac{B_t}{P_t}$ is real debt for a given individual borrower and the function ω is derived from banking intermediation in the section below. The variable M_t denotes aggregate private nominal debt in the economy, which the agents take as exogenous to their private decisions. The function ω is increasing in $b_t \equiv \frac{B_t}{P_t}$ and $m_t \equiv \frac{M_t}{P_t}$, i.e. $\omega_b > 0$ and $\omega_m > 0$ but we make the simplifying assumption that $\omega_{bm} = 0$. While all depositor face the same deposit rate, i_t^d, the rate each borrower faces, i_t^b, depends on each borrowers individual level of debt. While the level of debt will be the same across borrowers in equilibrium, this assumption affects the first order conditions derived below. The shock we consider is an unexpected drop in the value of b^h to b^l and once again we are interpreting this as the "safe" level of borrowing. A drop from b^h to b^l – assuming debt was as its former steady state – will then trigger a rise in borrowing costs until individual and aggregate debt come down to the new "safe" level b^l.

The Lagrangians each households solve are

$$\begin{aligned}
\mathcal{L}_t^b \quad = \quad & E_0 \sum_{t=0}^{\infty} \delta^t [u^b(C_t^b) - v^b\left(h_t^b\right) \\
& + \lambda^b \{B_t - (1 + i_{t-1}^d)(1 + \omega(\frac{B_{t-1}}{P_{t-1}}, \frac{M_{t-1}}{P_{t-1}}, b^{H,L}))B_{t-1} + P_t C_t^b - P_t W_t h_t^b \\
& + \int I_t^b(i)di + \int Z^b(i)di + P_t F_t^b + P_t T_t^b\}]
\end{aligned}$$

and

$$\begin{aligned}\mathcal{L}_t^b &= E_0 \sum_{t=0}^{\infty} \beta^t [u^s(C_t^s) - v^s(h_t^s) \\ &+\lambda_t^s \{D_t - (1+i_t^d)D_{t-1} + P_t C_t^s - P_t W_t h_t^s - \int I_t^s(i)di - \int Z_t^s(i)di - P_t F_t^s - P_t T_t^s\}]\end{aligned}$$

There are now two consumption Euler equations that are derived by combining the FOC of these Lagrangians:

$$u_c^b(C_t^b) = \delta E_t \frac{1+i_t^b}{\Pi_{t+1}} u_c^b(C_{t+1}^b) + \delta E_t u_c^b(C_{t+1}^b) \frac{B_t}{P_t} \frac{1+i_t^b}{1+\omega_t} \frac{\omega_b(\frac{B_t}{P_t}, \frac{M_t}{P_t}, b^j)}{\Pi_{t+1}} \tag{28}$$

$$u_c^s(C_t^s) = \beta E_t \frac{1+i_t^d}{\Pi_{T+1}} u_c^s(C_{t+1}^s) \tag{29}$$

The optimal labor supply for each type of household can also be found by manipulating the FOC of the Lagrangian and are given by

$$W_t = \frac{v_h^b(h_t^b)}{u_c^b(C_t^b)} \tag{30}$$

$$W_t = \frac{v_h^s(h_t^s)}{u_c^s(C_t^s)} \tag{31}$$

3.2 Firms

The C_t of each type refers to the Dixit-Stiglitz aggregator of preference for goods varieties

$$C_t \equiv [\int_0^1 c_t(j)^{\frac{\theta-1}{\theta}} dj]^{\frac{\theta}{\theta-1}}$$

where $c_t(j)$ refers consumption of variety j. There is a continuum of firms of measure one, each one producing each variety of the good. This specification of preferences implies that each firm faces a demand of the following form

$$y_t(j) = (\frac{p_t(j)}{P_t})^{-\theta} Y_t$$

The firm maximize profits given by

$$Z_t(i) = (1-\tau) p_t(i) y_t(i) - P_t W_t h_t(i)$$

We assume as in Calvo (1983), that firms only get to revisit their price with a probability $(1-\alpha)$ in each period, independent of their history. It can be shown that in that case all firms that get to pick their price will choose the same price given by p_t^*. We assume that the firms are controlled by the savers. The problem that each firm faces at the time it revises its price is then to choose a price p_t^* to maximize

$$\max_{p_t^*} E_t \left\{ \sum_{T=t}^{\infty} (\alpha\beta)^{T-t} \lambda_T^s [(1-\tau) p_t^* Y_T (p_t^*/P_T)^{-\theta} - P_T W_T Y_T (p_t^*/P_T)^{-\theta}] \right\}.$$

The first order condition of this problem implies an optimal price $\frac{p_t^*}{P_t}$ given by

$$\frac{p_t^*}{P_t} = \frac{E_t \sum_{T=t}^{\infty} (\alpha\beta)^{T-t} u_c^s(C_T^s) (\frac{P_T}{P_t})^{-\theta} Y_T \frac{\theta}{\theta-1} W_T}{E_t \sum_{T=t}^{\infty} (\alpha\beta)^{T-t} u_c^s(C_T^s) (1-\tau) (\frac{P_T}{P_t})^{-\theta-1} Y_T} \tag{32}$$

An important element to close the model is to observe that the Calvo assumption (that a $1-\alpha$ measure of firms choose p_t^* each period while the rest leaves their price unchanged) implies that

$$P_t = [(1-\alpha)(p_t^*)^{1-\theta} + \alpha P_{t-1}^{1-\theta}]^{1/(1-\theta)}$$

or equivalently

$$(\frac{p_t^*}{P_t}) = [\frac{1-\alpha\Pi_t^{-1+\theta}}{1-\alpha}]^{\frac{1}{1-\theta}} \tag{33}$$

For the aggregation of the model, in the section below, it will be helpful to define price dispersion as $\Delta_t \equiv \int (\frac{p_t(i)}{P_t})^{-\theta} di$. The Calvo assumption then implies that

$$\Delta_t = (1-\alpha)(\frac{p_t^*}{P_t})^{-\theta} + \alpha\Delta_{t-1}\Pi_t^{\theta} \tag{34}$$

3.3 Aggregation

The aggregate resource constraint is

$$Y_t = \chi_b C_t^b + \chi_s C_t^s + G_t \tag{35}$$

where G_t is government spending. Meanwhile, aggregate labor is

$$h_t = \chi_s h_t^s + \chi_b h_t^b = \int_0^1 y_t(i) = \int Y_t(\frac{p_t(i)}{P_t})^{-\theta} di = Y_t\Delta_t \tag{36}$$

where h_t is aggregate labor supply and note that we substituted in for the demand for varieties implied by the Dixit-Stiglitz preferences and the definition of price dispersion that we defined above.

Finally, the evolution of the real value of aggregate debt, $b_t \equiv \frac{B_t}{P_t}$, is

$$b_t = (1+i_{t-1}^b)b_{t-1}\Pi_t^{-1} + C_t^b - W_t h_t^b + T_t^b \tag{37}$$

where we have *assumed* that all the firms, banks and fraud profits are distributed to the savers. Note that this is a somewhat arbitrary assumption. We can easily change it but it seems natural that people that are borrowing at a premium do not hold any "stocks". To close the model we need a specification for monetary policy. It is a reaction function (or "Taylor rule")

$$i_t^d = \max(0, f(X_t)). \tag{38}$$

where the vector X_t denotes all the endogenous and exogenous variables in the economy (in the case of a Talyor rule it would refer to only output and inflation). We are not specific about the policy rule at the moment, and leave that to further discussions later on. And this closes the model so we can now define an equilibrium in the nonlinear model.

3.4 Equilibrium definition and a steady state in the non-linear model

We are now are in a position of defining an equilibrium. It is defined after the realization of an unexpected shock b^h to b^l. An equilibrium is a set of 13 deterministic processes $\{i_t^d, i_t^b, \omega_t, b_t, C_t^b, C_t^s, h_t^b, h_t^s, Y_t, W_t, \frac{p_t^*}{P_t}, \Pi_t, \Delta_t\}$ that solve the 13 equations (26)-(38). These equations have the following steady state that we will then linearize around. We study monetary and fiscal policy that implements a steady state in which $\Pi_t = 1$ and $\frac{p_t^*}{P_t} = 1$, $i^d = \beta^{-1} - 1$. The consumption Euler equation of the borrower, and the relationship between i^d and i^b pins down steady state debt as the $\bar{b} = \bar{m}$ that solves

$$\delta^{-1}\beta = 1 + \omega(\bar{b}, \bar{b}) + \bar{b}\omega_b(\bar{b}, \bar{b}) \tag{39}$$

And now, $\bar{i}^b = \beta^{-1}(1+\omega) - 1$ and $W = (1-\tau)\frac{\theta-1}{\theta} = 1$ (this is an assumption of a subsidy in steady state but this is not really needed but simplifies the notation a little bit) and $G = \bar{G}$. We choose the normalization $Y = 1$. C^b, C^s, h^b and h^s are then the constants $(\bar{C}^b, \bar{C}^s, \bar{h}^b, \bar{h}^s)$ that solve the following four equations

$$\bar{h}^b - \bar{C}^b = \bar{i}^b \bar{b} + G$$

$$\frac{v_h^b(\bar{h}^b)}{u_c^b(\bar{C}^b)} = 1$$

which determine $\bar{h}^b$ and $\bar{C}^b$ and then we can back out $\bar{h}^s$ and $\bar{C}^s$ from the following equations

$$\chi_s \bar{h}^s + \chi_b \bar{h}^b = \chi_b \bar{C}^b + \chi_s \bar{C}^s + \bar{G}$$

$$\frac{v_h^s(\bar{h}^s)}{u_c^s(\bar{C}^s)} = 1$$

Observe that we are linearizing around the steady state associated with $\bar{b} = b^l$ but we will be considering an unexpected decline in the "safe" level of debt from b^h to b^l. The initial value of the debt will thus be away from it's steady state, and this is the "impulse" of the model.

We are now ready to linearize. But before going into that lets quickly flesh out the how we got to the function $\omega(.)$ that determined the spreads by introducing a banking sector more explicitly.

3.5 Banking

We just stated in reduced from how spreads came about through the function $\omega(.)$. Now lets get into some details about financial intermediation to derive what we already assumed (i.e. the function $\omega(.)$), before moving to the linearized model.

There are banks that accept deposits $D_t(j)$ from some anonymous depositor and lends $B_t(i)$ to individual i The deposits are paid out the next period and the loan collected. Let us consider a "bank" as the life of a single loan contract. The banks profit as a result of this contract is

$$PR_t = \frac{D_t(j)}{P_t} - \frac{B_t(i)}{P_t} - \Upsilon_t(\frac{B_t(i)}{P_t}, \frac{M}{P_t}, b^i) + E_t \Gamma_{t,t+1}[(1+i_t^b)\frac{B_t(i)}{P_{t+1}} - (1+i_t^d)\frac{D_t(j)}{P_{t+1}}]$$

where $\Gamma_{t,t+1}$ is a stochastic discount factor and Υ_t is costs associated with financial intermediation. It is increasing in the level of debt away from the "safe" level of b^i that is tied to borrower i. The interpretation of the intermediation cost function is that Υ_t corresponds to funds that are "stolen" away, by "bad borrowers" posing as good borrowers and not paying back. For simplicity we suppose that Υ_t gets simply transferred to the savers (so the interpretation here is that the savers have an opportunity to "pose" as borrowers and abscond with the money). We assume that the function Υ_t is increasing in $\frac{B_t(i)}{P_t}$ away from b^i, but also increasing in aggregate lending $\frac{M_t}{P_t}$. The generalization of including aggregate private debt into $\Upsilon_t(.)$ is inessential but helps in that it allows us to consider a bigger parameter space in our numerical example. This clarifies the mechanism of the model a little bit as we will see and separates out different channels.

A key simplification assumption is the following:[1] To the extent there are profits of this loan contract, the bank pays the profit to its owner in period t and only holds enough assets to pay off the depositors in period $t+1$. This assumption implies that $(1+i_t^b)B_t(i) = (1+i_t^d)D_t(j)$ because the bank will only hold

[1] This is a nice trick we borrow from Curdia and Woodford (2009).

enough assets (through its perfect predictable outstanding loan $B_t(i)$) to pay off its depositor j in period $t+1$. Substituting this relationship we can write profits as

$$PR_t = \frac{1+i_t^b}{1+i_t^d}\frac{B_t(i)}{P_t} - \frac{B_t(i)}{P_t} - \Upsilon_t(\frac{B_t(i)}{P_t}, .)$$

and maximizing profits by the optimal choice of $B_t(i)$ yields

$$\frac{1+i_t^b}{1+i_t^d} - 1 - \Upsilon_1(.) = 0$$

or

$$\frac{i_t^b - i_t^d}{1+i_t^d} = \Upsilon_1(.)$$

Let us define $\omega_t \equiv \frac{i_t^b - i_t^d}{1+i_t^d}$ so that we now have that

$$1 + i_t^b = (1+i_t^d)(1+\omega_t)$$

and we have just shown that the spread is just the function $\omega_t = \omega(\frac{B_t(i)}{P_t}, \frac{M}{P_t}, b^j) = \Upsilon_1(.)$.

3.6 Linearization

Lets now move to the linearization. Equation (26) is

$$\hat{i}_t^b = \hat{i}_t^d + \hat{\omega}_t$$

where $\hat{i}_t^b \equiv \frac{i_t^b - i^b}{1+i^b}, \hat{i}_t^d \equiv \frac{i_t^d - i^d}{1+i^d}$, $\hat{\omega}_t \equiv \frac{\omega_t - \omega}{1+\omega}$ while (27) is

$$\hat{\omega}_t = \vartheta \hat{b}_t$$

where $\vartheta \equiv \frac{\bar{\omega}_b + \omega_m}{1+\bar{\omega}}\bar{b}$ and $\hat{b}_t \equiv \frac{b_t - \bar{b}}{\bar{b}}$. The two consumption Euler equations (28) and (29) are

$$\hat{C}_t^b = E_t\hat{C}_{t+1}^b - \sigma_b(\hat{i}_t^b - E_t\pi_{t+1} + \lambda\hat{\omega}_t)$$

$$\hat{C}_t^s = E_t\hat{C}_{t+1}^s - \sigma_s(\hat{i}_t^d - E_t\pi_{t+1})$$

where $\lambda \equiv \delta\beta^{-1}(1+\bar{\omega}) - \delta\beta^{-1}\bar{b}\bar{\omega}_b + \delta\beta^{-1}\bar{b}\frac{\bar{\omega}_{bb}}{\bar{\omega}_b}(1+\bar{\omega}), \sigma_b = \sigma_s = -\frac{u_{cc}^b Y}{u_c^b} = -\frac{u_{cc}^s Y}{u_c^s}$ and $\hat{C}_t^b \equiv \frac{C_t^b - \bar{C}^b}{\bar{Y}}$, $\hat{C}_t^s \equiv \frac{C_t^s - \bar{C}^s}{\bar{Y}}$ and $\pi_t \equiv \frac{P_t}{P_{t-1}} - 1$

Meanwhile the optimal labor supply of each type is

$$\hat{W}_t = \nu\hat{h}_t^b + \sigma_b^{-1}C_t^b$$

$$\hat{W}_t = \nu\hat{h}_t^s + \sigma_s^{-1}\hat{C}_t^s$$

where $\nu \equiv \frac{v_h}{v_{hh}Y}$. Equation (32) and (34) imply that

$$\hat{\Delta}_t = 0$$

and

$$\hat{p}_t^* = \frac{\alpha}{1-\alpha}\pi_t$$

where $\hat{p}_t^* \equiv \frac{p_t^*}{P_t} - 1$.

This and (32) can be used to show that (the step here are the standard ones used to derive the New Keynesian Phillips curve, see, e.g. Woodford (2003), chapter 3)

$$\pi_t = \gamma \hat{W}_t + \beta E_t \pi_{t+1}$$

where $\gamma \equiv \frac{(1-\alpha)(1-\alpha\beta)}{\alpha}$.

The aggregate resource constraint (35) is

$$\hat{Y}_t = \chi_b \hat{C}_t^b + \chi_s \hat{C}_t^s + \hat{G}_t$$

where $\hat{G}_t \equiv \frac{G_t - \bar{G}}{\bar{Y}}$ and $\hat{Y}_t \equiv \frac{Y_t - \bar{Y}}{\bar{Y}}$.

Meanwhile the labor resource constraint (36) yields (using our result $\hat{\Delta}_t = 0$)

$$\hat{Y}_t = \chi_b \hat{h}_t^b + \chi_s \hat{h}_t^s \tag{40}$$

The budget constraint of the borrower 37 is

$$b_y \hat{b}_t = b_y(1+\bar{\imath}^b)\hat{\imath}_{t-1}^b + (1+\bar{\imath}^b) b_y \hat{b}_{t-1} - b_y(1+\bar{\imath}^b)\pi_t + \hat{C}_t^b - h^b \hat{W}_t - \hat{h}_t^b + \hat{T}_t^b$$

where $b_y \equiv \frac{b}{Y}$, $h^b \equiv \frac{h^b}{h}$, $\hat{T}_t^b \equiv \frac{T_t^b - T^b}{Y}$ and we have used that in steady state $W = Y = h = 1$. Finally the monetary policy rule is

$$\hat{\imath}_t^d = \max(\beta - 1, \phi_x \hat{X}_t)$$

where we have used that the zero bound, $i_t^d \geq 0$ implies that $\hat{\imath}_t^d \geq -\frac{i^d}{1+i^d} = \beta - 1$. The policy rule is written in a "general way" as $\hat{X}_t$ is the vector of all endogenous and exogenous variables, and ϕ_x is a vector of numbers of coefficients. Below, we will actually only consider a policy that targets zero inflation whenever the zero bound is not binding (there are many ways of implementing this sort of policy with an appropriate choice of ϕ_x given a particular equilibrium selection device).

3.7 Some manipulations and interpretations

The equations are summarized below. Then some interpretations in the next subsection

3.7.1 Summary of equations

$$\hat{\imath}_t^b = \hat{\imath}_t^d + \hat{\omega}_t \tag{41}$$

$$\hat{\omega}_t = \vartheta \hat{b}_t \tag{42}$$

$$\hat{C}_t^b = \hat{C}_{t+1}^b - \sigma(\hat{\imath}_t^b - E_t \pi_{t+1} + \lambda \hat{\omega}_t) \tag{43}$$

$$\hat{C}_t^s = E_t \hat{C}_{t+1}^s - \sigma(i_t^d - E_t \pi_{t+1}) \tag{44}$$

$$\hat{W}_t = \nu \hat{h}_t^b + \sigma^{-1} \hat{C}_t^b \tag{45}$$

$$\hat{W}_t = \nu \hat{h}_t^s + \sigma_s^{-1} \hat{C}_t^s \tag{46}$$

$$\pi_t = \gamma \hat{W}_t + \beta E_t \pi_{t+1} \tag{47}$$

$$b_t = b_y(1+\bar{\imath}^b)\hat{\imath}_{t-1}^b + (1+\bar{\imath}^b)\hat{b}_{t-1} - b_y(1+\bar{\imath}^b)\pi_t + \hat{C}_t^b - h^b \hat{W}_t - \hat{h}_t^b + \hat{T}_t^b \tag{48}$$

$$\hat{Y}_t = \chi_b \hat{C}_t^b + \chi_s \hat{C}_t^s + \hat{G}_t \tag{49}$$

$$\hat{Y}_t = \chi_b \hat{h}_t^b + \chi_s \hat{h}_t^s \tag{50}$$

$$\hat{\imath}_t^d = \max(\beta - 1, \phi_x \hat{X}_t) \tag{51}$$

so we have 11 equation in 11 unknowns $\{i_t^b, i_t^b, \omega_t, b_t, C_t^b, C_t^s, h_t^b, h_t^s, Y_t, \pi_t, W_t\}$. Note that the "initial" shock here is an unexpected decline in the "safe" borrowing limit, b^h to b^l. This corresponds to a large debt "overhang" as an initial value, i.e. $\hat{b}_{-1} > 0$ for $t \geq 0$ that needs to be paid down, but we are linearizing around the "new" safe level of debt. Again observe that we have not been specific about the policy rule but we will be so soon. In the simulations below we are going to assume that the central bank targets zero inflation if it can.

3.7.2 An interpretation and connection with standard New Keynesian model

If we look a bit closer it becomes clear that this is really just the standard New Keynesian model with a twist. In particular, lets write the model in terms of $\{i_t^d, \omega_t, Y_t\}$ as is standard (lets ignore government spending for a moment, but it shows up in the same way as in the standard model). Then with just a little bit of work we can derive

$$\hat{Y}_t = E_t\hat{Y}_{t+1} - \sigma(\hat{i}_t^d - E_t\pi_{t+1} + \chi_b(1+\lambda)\hat{\omega}_t) \tag{52}$$

$$\pi_t = \kappa\hat{Y}_t + \beta E_t\pi_{t+1} \tag{53}$$

$$\hat{i}_t^d = \max(\beta - 1, \phi_\pi\pi_t + \phi_x\hat{Y}_t) \tag{54}$$

This is simply the "textbook" New Keynesian model, e.g. Woodford (2003) and Gali (2008), with only one twist. Usually the last term in the first equation (the IS equation) is written as $-\hat{r}_t^e$, i.e. the natural rate of interest is usually exogenous (and then we have three equation in three unknown and the model can be solved).

But now the "natural rate of interest" is *endogenous*, i.e. $\hat{r}_t^e = -\chi_b(1+\lambda)\hat{\omega}_t = -\chi_b(1+\lambda)\vartheta\hat{b}_t$, so we need to figure out the evolution of debt of the "borrowers" to figure out the natural rate of interest. In particular we see that if $\hat{b}_t$ is high (i.e. the economy is "overleveraged") it is easy to get endogenously *negative* natural rate of interest. And then we need to figure out the endogenous evolution of all the variables in the system summarized in the last section.

Also not that here we have exactly the same story as in Eggertsson and Krugman (2010). The debt overhang, i.e. a positive $\hat{b}_{-1}$ initial value, means that is going to be an increase in the "spreads" $\hat{\omega}_t$. This means that the borrower will deleverage. But somebody needs to make up for this spending, and this falls on the saver. But the only way he can do that is if real interest rate declines. And for a large enough debt overhang the real interest rate can be negative. This will spell trouble.

3.8 Numerical example

3.8.1 Parameterization

parameter	value	parameter	value
β	0.995	b_y	4
δ	0.96	χ	0.5
κ	0.02	$1 + i^b$	1.02
ν	1	ϑ	0.049
σ	1	λ	1
		$\hat{b}_{-1}$	0.3

Table 1

For the numerical experiment we choose numbers that are standard in the literature and then move on to the parameters that are new as we explain below. Each period is a quarter. Start with parameters that are familiar in the literature, summarized in the left hand side in the table. We assume $\beta = 0.995$ to match

a 2 percent risk-free real interest rate. We assume that $\sigma_b = \sigma_s = 1$ and similarly that $\nu_s = \nu_b = 1$ – both values are common in the literature. We assume that κ takes on the value 0.02, as in Woodford (2003), but this composite parameter is a function of several "structural parameters" (see derivation in past sections).

Lets now turn to the right hand side of the table which is less standard. Start with steady state debt. We assume $b^l = \bar{b} = 4$, i.e. steady state debt is 100 percent of annual income (recall that each period is a quarter). For the initial shock we choose $\hat{b}_{-1} = 0.3$ i.e. the initial value of the debt is 30 percent above steady state (Both numbers are motivated by the Table 1 in Eggertsson and Krugman (2010)). We assume that each type is of equal proportion, i.e. $\chi = 0.5$. We assume the steady state interest rate faced by borrowers is 8 percent per annum, so that the spread between i^d and i^b is 6 percent (we may want to get some data on this). This pins down $1 + \bar{\omega} = \frac{1+i^b}{1+i^d}$. Finally we come to two composite parameters, λ and ϑ which are what remains to choose values for the linearized system (41)-(51). These parameters can take on any positive value, which will be a function of the parameters $(\bar{\omega}_{bb}, \bar{\omega}_b, \bar{\omega}_m, \delta)$ but these four parameters have to satisfy the steady state relationship $\delta^{-1}\beta = 1 + \bar{\omega} + \bar{b}\bar{\omega}_b$. For a given value of λ and ϑ (and the parameters in the table) there are many possible $(\bar{\omega}_{bb}, \bar{\omega}_b, \bar{\omega}_m, \delta)$, i.e. these four parameters are not "identified". Below we simply directly choose λ and ϑ and then discuss how they depend on the structural parameters and consider some special cases.

Let us start with ϑ. Observe that according to 42 we have that $\hat{\omega}_t = \vartheta \hat{b}_t$ so that for a given initial value of debt, we can roughly target the spread, $\hat{\omega}_t$, by the choice of ϑ.[2] We choose ϑ so that the spread would double if the value of debt is given by b_{-1}, i.e. $\vartheta = \frac{\bar{\omega}}{1+\bar{\omega}} \frac{1}{b_{-1}} = 0.0489$. Lets now turn to λ which is last parameter needed to fully characterize the model. This parameter must be greater than or equal to zero. If it is exactly zero, this implies that ω_b is also zero, and aggregate debt is the only thing that matters for spreads. In this special case the borrower will not try to deleverage for the purpose of reducing his own borrowing cost (this is the same assumption as in Curdia and Woodford (2009)) as the spread depend only on aggregate debt (and each agent is small). The parameter λ, however, can very well be positive if the borrowers individual debt has an effect on the terms at which he can borrow. In particular we will see that as λ is increased then once the zero bound becomes binding, the recession becomes worse and worse, due to the fact that the borrower will then start cutting spending by more and more, i.e. deleverage faster. In fact the forces at work here are so strong that at some critical value $\bar{\lambda}$ the drop in aggregate output is unbounded, i.e. the model explodes. Here we simply pick $\lambda \in [0, \bar{\lambda}]$ such that that drop in output corresponds to what we saw in the Great Recession – i.e. and output drop of about 7 percent from steady state — which results in $\lambda = 1.32$. [3] We discuss this strategy in more detail in the zero bound numerical example. But before getting there, it is useful to understand the basic properties of the model abstracting from the zero bound.

3.8.2 Deleveraging shock under flexible prices (or zero inflation target)

Consider first what would happen if prices were flexible. An alternative interpretation of this equilibrium is that it corresponds to model with price frictions but under the assumption that the central bank targets zero inflation and the zero interest rate bound is never a problem. The reason this interpretations works is that the deleveraging shock, as one can see by inspecting equations 52 and 53, creates no trade-off between inflation and output in the absence of the zero bound. A central bank that successfully targets zero inflation, then, is simply replicating the flexible price allocation.

Figure 3 and 4 shows what happens for a large deleveraging shock if the central bank successfully

[2] We say roughtly, because the state variable is b_{t-1} – not b_t – so the mapping is not one to one.

[3] Again, note that these values of ϑ and λ, do not in general pin down $(\bar{\omega}_{bb}, \bar{\omega}_b, \bar{\omega}_m, \bar{\delta})$, but if one were to take a stance of one of these four parameters, then the others can be backed out of the steady state relationships for λ and ϑ and $\delta^{-1}\beta = 1+\omega+\bar{b}\omega_b$.

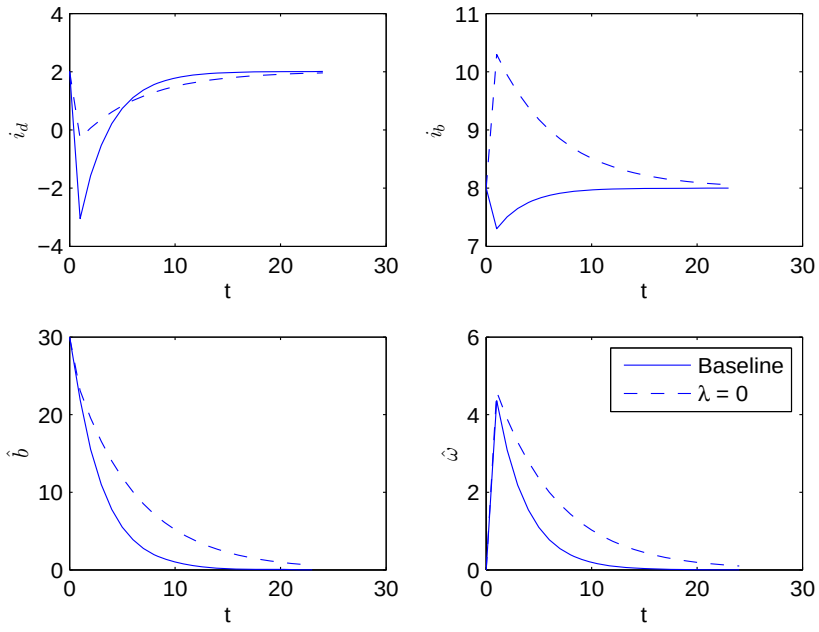


Figure 3: Risk-free nominal rate, risky-nominal rate, private debt and the spread under a zero inflation target abstracting from the zero bound.

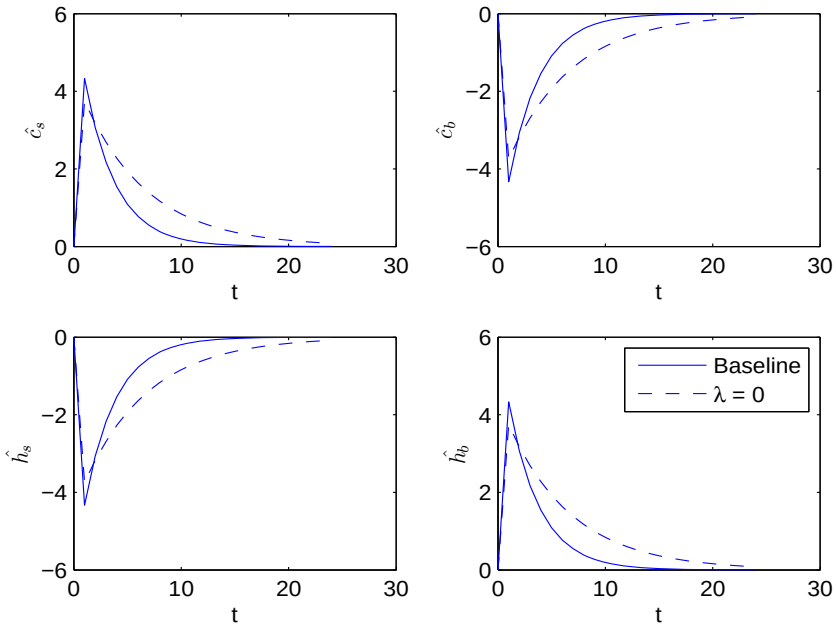


Figure 4: Consumption and hours under a zero inflation target abstracting from the zero bound.

targets zero inflation. The "deleveraging shock" reflects the consequence of the steady state "safe debt level" b^h goes unexpectedly to b^l. What this means in the approximate model is that we are considering the behavior of the model with an initial level of debt that is above steady state, i.e. $b_{-1} > 0$, as an initial conditions. As stated in the last section, in our numerical experiment we consider a shock so that private debt is 30 percent above steady state. The benchmark parameterization is shown with a solid line.

Figure 3 shows that a debt of this order is paid down over a horizon of about 15 quarters. The "borrower" deleverages in this way by both cutting consumption down by about four percent and increasing hours worked by about the same amount. This is offset by a symmetric behavior by the saver. The net effect on output is zero and there is also no effect on prices.

To achieve this equilibrium the central bank, which controls the risk-free rate i^d, cut's it to a negative level, and as figure 41 shows the rates stay negative for about four quarters. Meanwhile, the borrower is facing a slightly higher borrowing rate (in nominal terms), so the spread between the deposit and borrowing rate is increasing as shown in the fourth panel in figure 41. The reason the shocks triggers an increase in spreads is that the banks now needs to be compensated more for the loans they extend according to the function ω, due to aggregate pressures and/or that the "save" level of lending to each borrower is different from before.

Figure 3 and 4 also show a variation on the baseline calibration when the central bank successfully targets zero inflation. In the baseline we assumed that $\lambda = 1.32$. This means that the borrower can reduce his cost of funding by bringing down is own debt. In comparison the dashed line show the case when $\lambda = 0$. In this case the only cost if borrowing is due to aggregate debt, and hence the borrower does not have as strong incentive to pay down his debt. In this case the agents will deleverage slower.

3.8.3 Deleveraging shock at zero interest rates and the effect of government spending

We now take into account that the central bank cannot set nominal interest rate below zero due to the zero bound. In other words, the central bank winds up in a liquidity trap. To solve the model we impose the zero bound directly, and look for a solution in which the zero bound is binding for the shortest period possible that is consistent with the model. The algorithm is implemented in Matlab by guessing the number of periods the zero bound is binding (starting with zero), and then checking if these guess is verified. The result is reported in figures 5 and 6.[4]

We see that the shock now triggers the zero bound to be binding for 10 periods, or two and a half years. The duration of the trap is determined by the endogenous deleveraging process. Note that relative to the case where the zero bound is not imposed, then the deleveraging process is slower. Meanwhile, output drops by about 7 percent, and inflation by about 2 percent, but both variables stayed at steady state before. The spreads roughly double – by construction since this is how we chosen ϑ – so that the risky borrowing rate increases to about 11 percent per annum from 8. The output collapse is triggered by the fact that the consumption of the borrowers collapses, and this fall is not offset by a large enough increase in consumption by the savers, which happened when the central bank could cut the risk-free rate to negative levels.

The dashed line shows the effect of an increase in government spending by 1 percent. We can see that the effect is very large, the output drop is basically eliminated. Moreover we see that the spending increase spreads up the deleveraging process considerably, so that now the zero bound is binding for less than a year, down from 2 and a half years. Thus it looks like spending here is "jump-starting the economy" as we mentioned in the into.

[4] We report the equilibrium which corresponds to the lowest number of periods in which the zoer bound is binding. We have not checked if other equilibria exist in which the zero bound is binding for a longer period of time.

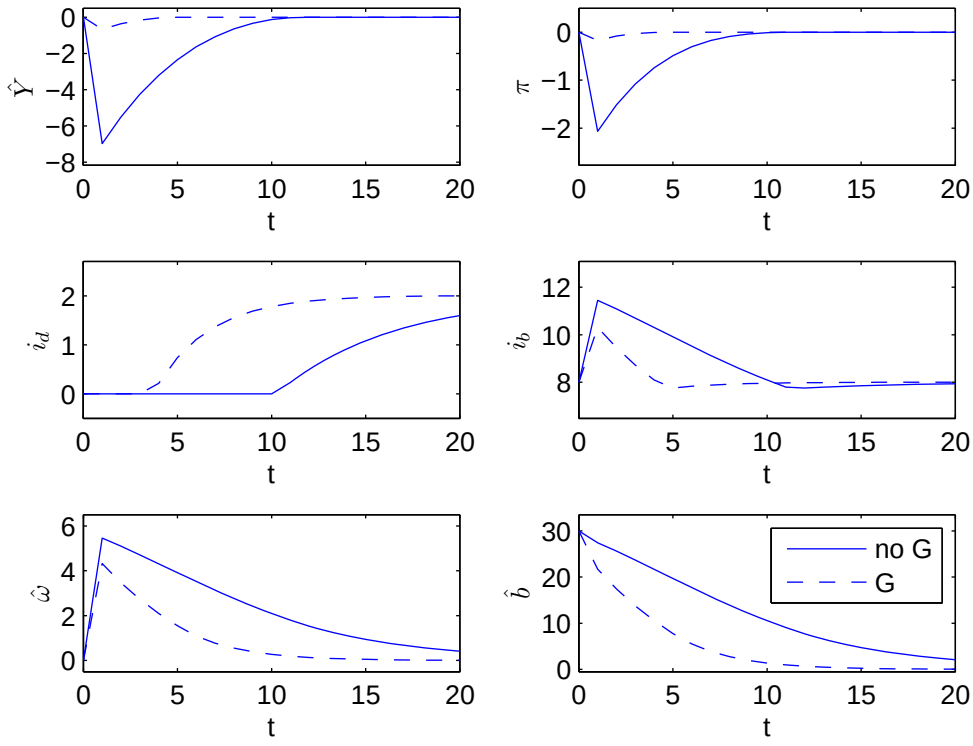


Figure 5: The figure shows the effect of a deleveraging shock once we take the zero bound into account, and how effective government spending is in counteracting it.

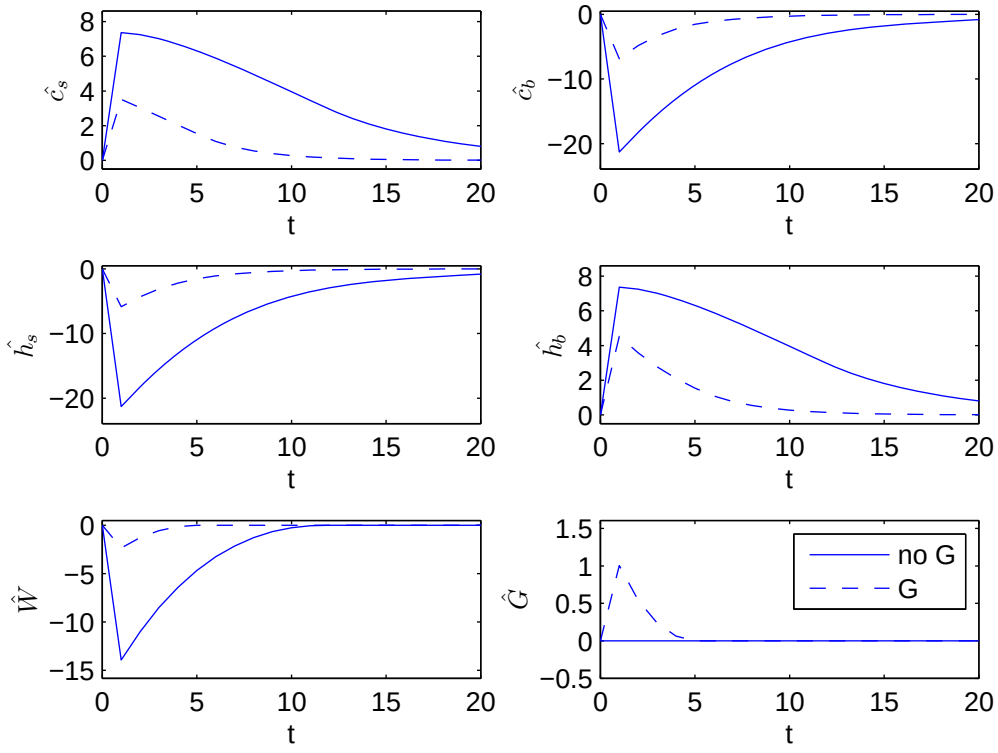


Figure 6: The figure shows the effect of a deleveraging shock once we take the zero bound into account, and how effective government spending is in counteracting it.

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